

# Monthly Intelligence Report

Edition 001 — The Osaka–Aichi Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

## CONTENTS

- [A](#) Executive Summary
- [B](#) Cyber & Economic Exposure Monitor
- [C](#) Data Refresh & Changelog
- [D](#) Deep-Dive: The Osaka–Aichi Corridor
- [E](#) International Peer Benchmarking Card
- [F](#) SSI Enhanced Neural Network
- [G](#) Looking Ahead

## SECTION A

### Executive Summary

#### SUBSTATIONS SCORED

5,981

299 EHV · 5,682 HV

#### GRID LINES MAPPED

~11,600

~105,000 km coverage

#### MC SIMULATIONS

78.9 M

10,000 per substation

#### PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 78.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 5,981 substations and ~11,600 grid lines spanning ~105,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Japan's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

### Substation in Focus

#### 一関変電所

Miyagi, Tohoku · 154 kV

0.607

High

0.307

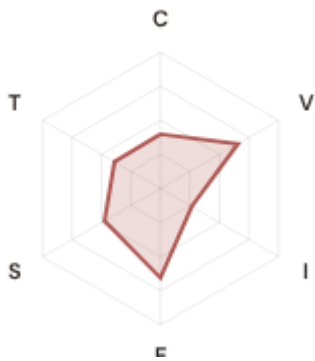
67th

R\_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



#### COMPONENTS

- C** Continuity: 0.399 / 0.30 (133%)
- I** Infrastructure: 0.265 / 0.25 (106%)
- S** Saturation: 0.474 / 0.20 (237%)
- V** Voltage: 0.657 / 0.10 (657%)
- E** Economic: 0.656 / 0.10 (656%)
- T** Transition: 0.390 / 0.05 (780%)

#### MODIFIERS

- R3 Consequence:** 1.055 ↑ amplifies
- R4 Graph Criticality:** 1.223 ↑ amplifies
- R6a Restoration:** 1.009 → neutral
- R6b Seismic:** 1.078 ↑ amplifies
- R7 Digital Readiness:** 1.001 → neutral

This month's spotlight — automatically selected for April 2026 — is 一関変電所 in Miyagi, Tohoku. With an R\_final of 0.607 (High band, 67th fleet percentile), its risk profile is dominated by the T (Transition) component at 780% of its weight ceiling, followed by V (Voltage) at 657%. Modifiers collectively shift R\_base by 40.6, reflecting the substation's specific geographic, socio-economic, and network context.

### SECTION B — RECURRING MODULE

## Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Society 5.0 / Japan 2030 digitalisation planning requires.

#### CYBER HIGH EXPOSURE

0

0.0% of fleet

#### BLIND SPOTS

2236

High/Critical + R7 < median

#### AVG R7 MODIFIER

1.0012

Range [0.99, 1.05]

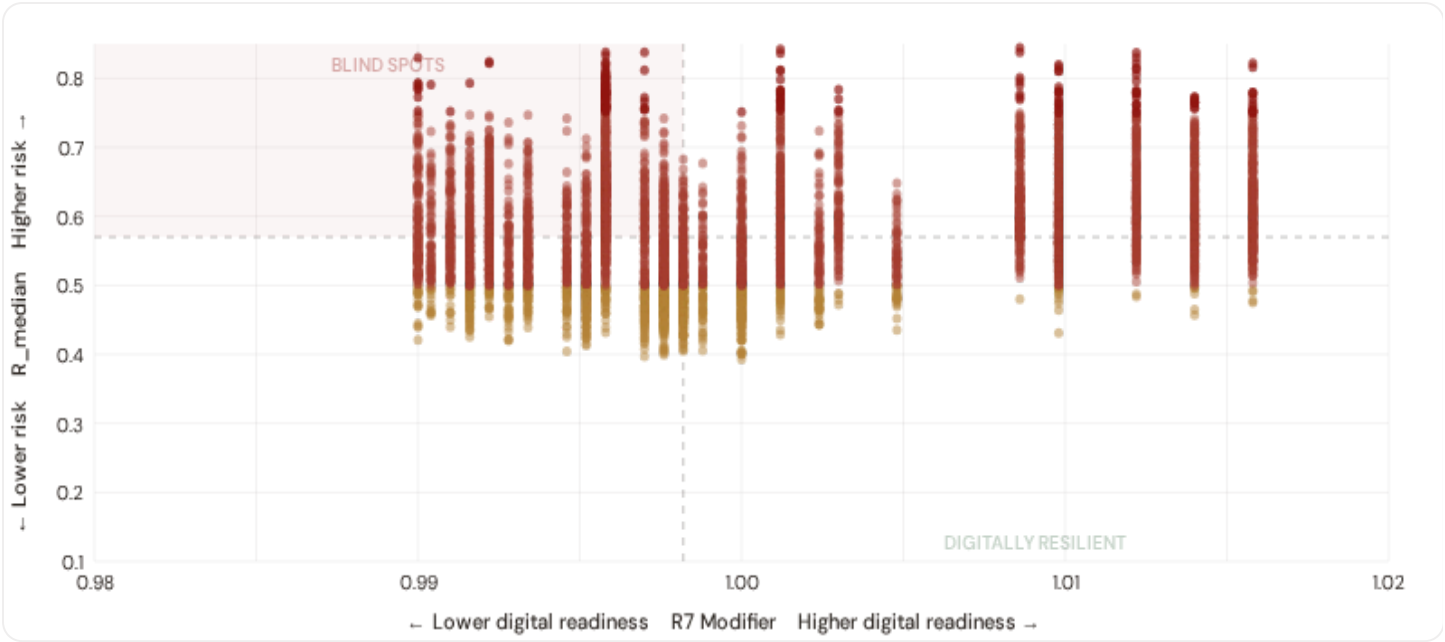
#### HIGH-CONSEQUENCE SUBS

2,431

40.6% · R3 ≥ 1.12

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R\_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

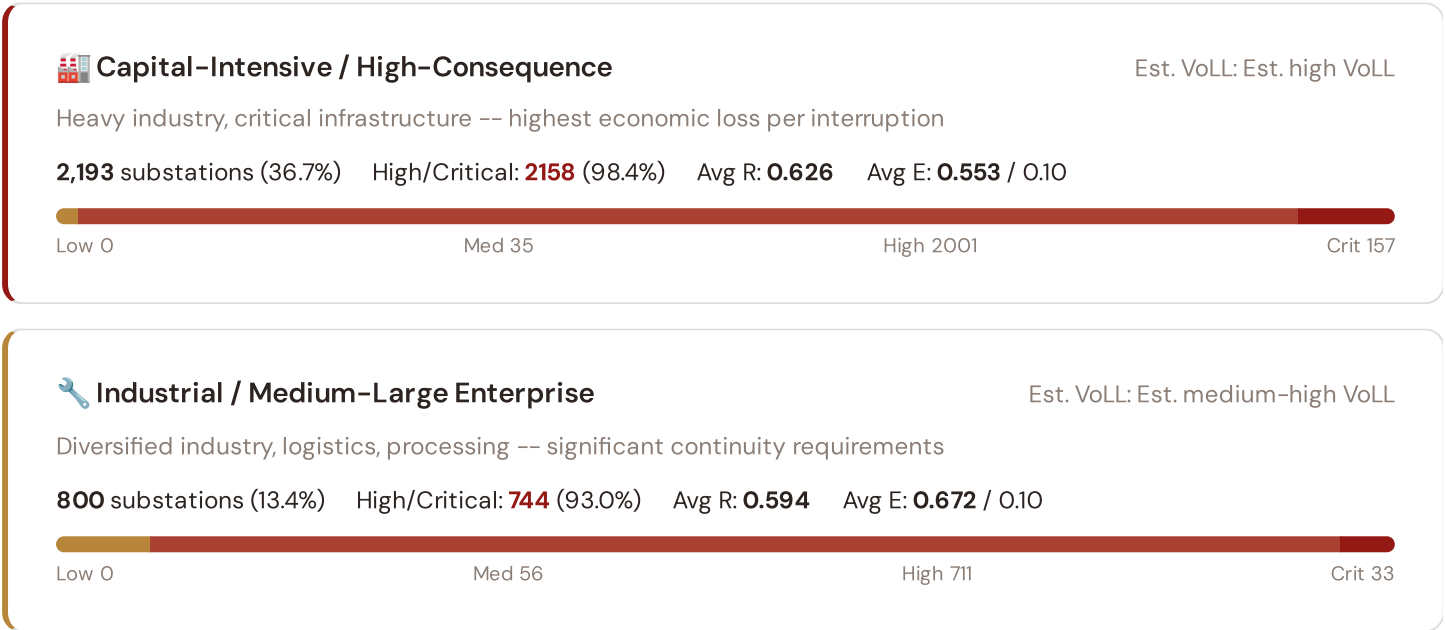


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **2236 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ( $R7 < 0.9982$ ). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Tohoku (686)**, **Hokkaido (403)**, **Kyushu (335)**. Across the full fleet, 0 (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the southern urban grid.

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.





### Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

1,533 substations (25.6%) High/Critical: **1165** (76.0%) Avg R: **0.551** Avg E: **0.689** / 0.10



### Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

1,455 substations (24.3%) High/Critical: **888** (61.0%) Avg R: **0.527** Avg E: **0.661** / 0.10

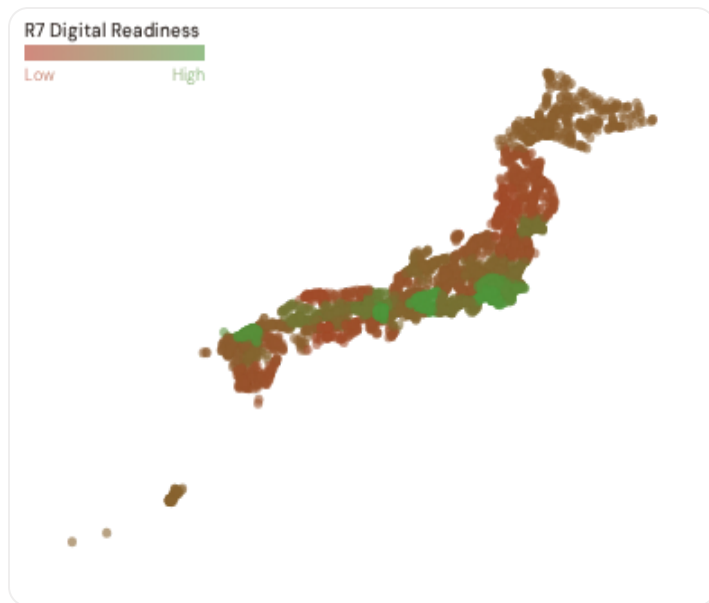


**Value of Lost Load (VoLL) context:** METI's 2024 estimate for Japan places average VoLL at ¥17/kWh for industrial customers and ¥4.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **2158 substations** combine capital-intensive economic fabric ( $R3 \geq 1.14$ ) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **TEPCO (1068), Hokkaido (403), Kansai (386)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of ¥20–40/kWh, compared to ¥1–4/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.0%, but this masks sharp regional variation: Remote and northern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 Prefecture Digital Readiness Ranking

Prefectures ranked by average R7 modifier — lower values indicate weaker digital infrastructure (MIC digital infrastructure sub-dimensions). Cross-referenced with average R<sub>median</sub> to identify prefectures combining digital exposure with high grid stress.



#### WORST DIGITAL READINESS — TOP 15 PREFECTURES

Longer bar = higher R7 = better digital infrastructure.

|    |                                 |             |        |
|----|---------------------------------|-------------|--------|
| 1  | Akita <span>▲ high R</span>     | <div></div> | 0.9900 |
| 2  | Kochi <span>▲ high R</span>     | <div></div> | 0.9900 |
| 3  | Shimane                         | <div></div> | 0.9900 |
| 4  | Yamagata                        | <div></div> | 0.9904 |
| 5  | Iwate                           | <div></div> | 0.9910 |
| 6  | Tokushima <span>▲ high R</span> | <div></div> | 0.9910 |
| 7  | Tottori                         | <div></div> | 0.9916 |
| 8  | Miyazaki <span>▲ high R</span>  | <div></div> | 0.9916 |
| 9  | Aomori                          | <div></div> | 0.9916 |
| 10 | Fukushima <span>▲ high R</span> | <div></div> | 0.9922 |
| 11 | Kagoshima <span>▲ high R</span> | <div></div> | 0.9922 |
| 12 | Wakayama <span>▲ high R</span>  | <div></div> | 0.9922 |
| 13 | Fukui                           | <div></div> | 0.9928 |
| 14 | Oita                            | <div></div> | 0.9928 |
| 15 | Niigata                         | <div></div> | 0.9934 |

Prefecture-level analysis reveals a clear digital divide mirroring the broader digital divide pattern. **Akita** has the lowest average R7 (0.9900), while **Tokyo** leads at 1.0218 — a spread of 0.0318 across the R7 range. **11 prefectures** combine below-median digital readiness with above-median grid risk, making them priority targets for Japan 2030 / OCCTO / OCCTO digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open MIC/METI digital data — as Cybersecurity Basic Act compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

## B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0012, median = 0.9982; 0 substations classified HIGH cyber-exposure; 2236 blind-spot substations (High/Critical risk + below-median R7); 89 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging prefectures where digital infrastructure improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when MIC publishes the 2025 MIC digital infrastructure indices (anticipated Q3 2026).

## SECTION C

## Data Refresh & Changelog

The SSI v4.0.2 draws on **32 verified public data sources** feeding 95 variables across 5 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

WEEKLY / MONTHLY

QUARTERLY / ANNUAL

**32**

95 variables

**20**

High-frequency feeds

**12**

Regular refresh cycle

STATIC / DERIVED

**0**

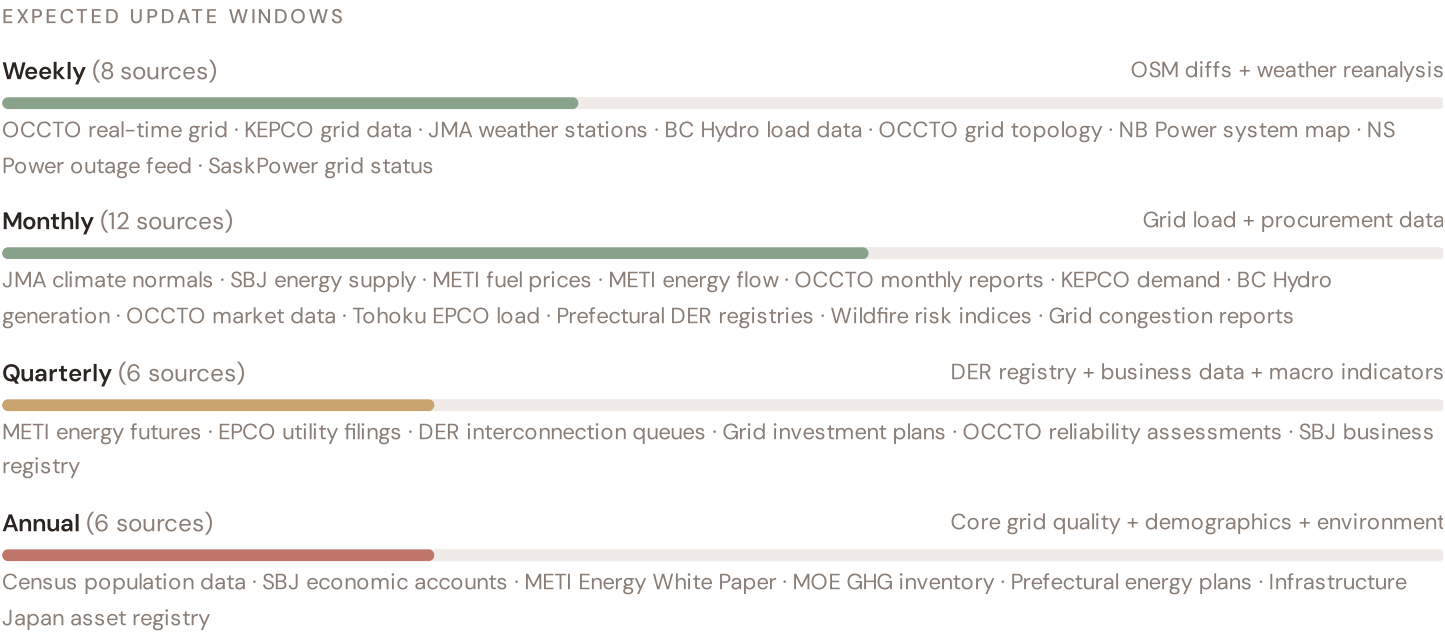
Standards + scenarios

Data Source Registry — April 2026

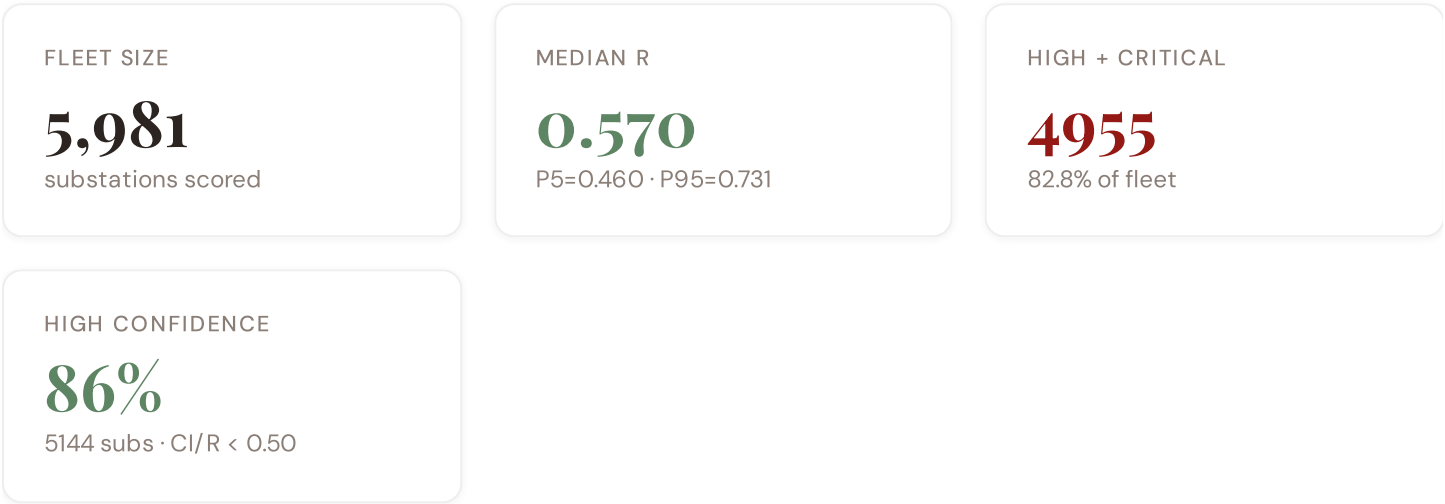
|               |                                                     |            |                |        |
|---------------|-----------------------------------------------------|------------|----------------|--------|
| OCCTO         | OCCTO Area Supply-Demand Statistics                 | MONTHLY    | EPCO Territory | 3 vars |
| OpenStreetMap | OpenStreetMap Overpass API — Japan Grid             | CONTINUOUS | Site-level     | 4 vars |
| JAXA          | JAXA Satellite-Derived Climate Data                 | QUARTERLY  | Regional       | 3 vars |
| BOJ           | BOJ Regional Economic Reports (Sakura Report)       | QUARTERLY  | Regional       | 3 vars |
| ISEP          | ISEP Energy Statistics                              | ANNUAL     | Prefecture     | 2 vars |
| METI          | METI Electricity Business Statistics                | ANNUAL     | EPCO Territory | 3 vars |
| SBJ           | SBJ Population Census / Current Population Survey   | ANNUAL     | Prefecture     | 4 vars |
| TEPCO         | TEPCO Grid Reliability Reports                      | ANNUAL     | EPCO Territory | 4 vars |
| Kansai        | Kansai Electric Reliability Data                    | ANNUAL     | EPCO Territory | 3 vars |
| Chubu         | Chubu Electric Power Reliability Reports            | ANNUAL     | EPCO Territory | 3 vars |
| Kyushu        | Kyushu Electric Power Reliability Data              | ANNUAL     | EPCO Territory | 2 vars |
| Hokkaido      | Hokkaido Electric Power Reports                     | ANNUAL     | EPCO Territory | 2 vars |
| Tohoku        | Tohoku Electric Power Data                          | ANNUAL     | EPCO Territory | 2 vars |
| Hokuriku      | Hokuriku / Shikoku / Chugoku / Okinawa EPCO Reports | ANNUAL     | EPCO Territory | 2 vars |
| NISC          | NISC Cybersecurity Reports                          | ANNUAL     | National       | 2 vars |
| MHLW          | MHLW Regional Health Statistics                     | ANNUAL     | Prefecture     | 3 vars |
| MIC           | MIC Digital Transformation Index                    | ANNUAL     | Prefecture     | 3 vars |
| Cabinet       | Cabinet Office GDP / Regional Economic Accounts     | ANNUAL     | Prefecture     | 3 vars |
| FDMA          | FDMA Fire and Disaster Management Statistics        | ANNUAL     | Prefecture     | 3 vars |
| METI          | METI Smart Grid Demonstration Data                  | ANNUAL     | EPCO Territory | 3 vars |
| MEXT          | MEXT University/Research Infrastructure Atlas       | ANNUAL     | Prefecture     | 2 vars |
| MIC           | MIC Migration Statistics (Jumin-Kihon-Daicho)       | ANNUAL     | Prefecture     | 2 vars |
| NPA           | NPA Crime Statistics and Security Reports           | ANNUAL     | Prefecture     | 1 var  |
| METI          | METI Equipment Age Survey / TEPCO Asset Reports     | ANNUAL     | EPCO Territory | 3 vars |
| GSI           | GSI Elevation and Terrain Data                      | STATIC     | 10m mesh       | 3 vars |
| ISO           | ISO 9223 / JIS Z 2381 Atmospheric Corrosion Maps    | STATIC     | Prefecture     | 1 var  |

|      |                                                 |                      |                   |        |
|------|-------------------------------------------------|----------------------|-------------------|--------|
| AIST | AIST Geological Survey Data                     | STATIC               | Regional          | 2 vars |
| IEEE | IEEE C57.91 Thermal Loading Standards           | STATIC               | Equipment-level   | 2 vars |
| IEC  | IEC 61850 / JEC Standards Compliance            | STATIC               | Equipment-level   | 2 vars |
| NIED | NIED J-SHIS Seismic Hazard Maps                 | UPDATED PERIODICALLY | 250m mesh         | 2 vars |
| JMA  | JMA Seismological Bulletin / Typhoon Statistics | REAL-TIME + ANNUAL   | National/Regional | 3 vars |
| MLIT | MLIT Flood and Landslide Hazard Maps            | UPDATED PERIODICALLY | Municipality      | 3 vars |
| JEPX | JEPX Wholesale Electricity Market Data          | DAILY                | Area              | 3 vars |
| NRA  | NRA Nuclear Safety Reports                      | AS NEEDED            | Site-level        | 3 vars |
| MIC  | MIC Housing and Land Survey                     | QUINQUENNIAL         | Prefecture        | 3 vars |

### C.1 Update Frequency Timeline



### C.2 Fleet Score Snapshot





BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **5,981 substation scores remain unchanged** this edition. The fleet median R of 0.570 (mean 0.578) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 17.2% in Medium. The first material score changes are expected when METI publishes the annual SAIDI/SAIFI indicators (affecting C1–C4 and R6a) and when OSM refreshes the METI Asset Registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across PO, P1, and P2 releases

|    |          |                                                                                        |           |
|----|----------|----------------------------------------------------------------------------------------|-----------|
| F1 | NEW      | New T component â Energy Transition Exposure (T1)                                    | \$2, \$4  |
| F2 | NEW      | New R6a â Restoration Speed Modifier (CAIDI-based)                                   | \$5       |
| F3 | ENHANCED | R3 enhanced â Energy Poverty Vulnerability (V_socio)                                 | \$5       |
| F4 | ENHANCED | R4 enhanced â Graph-theoretic betweenness + bridge detection                         | \$5       |
| F5 | ENHANCED | R2 enhanced â Climate Trajectory (CMIP6 SSP2-4.5)                                    | \$5       |
| F6 | ENHANCED | R7 Digital Readiness — Prefecture-level MIC digital infrastructure model               | \$5       |
| L1 | NEW      | New I5, I7–I9 metrics â thermal, load, corrosion, hydrogeo                           | \$2, \$8  |
| L2 | ENHANCED | E2 Innovation Enrichment — MEXT R&D + university density                               | \$6       |
| L3 | ENHANCED | V_socio Fiscal Enrichment — MIC + MOF + energy price indices                           | \$5       |
| L4 | ENHANCED | R3 Demographic Shift Amplifier — SBJ net migration                                     | \$5       |
| L5 | DATA     | 95/95 variables operational (100%). 35 data sources total.                             | \$8, \$12 |
| G1 | DATA     | d02 ISEP upgraded to quarterly registry — Prefecture-level DER registry                | \$12      |
| G2 | DATA     | d05 AIST/GSJ upgraded to live API — Prefecture-level geological data                   | \$12      |
| G3 | DATA     | d06 OSM upgraded to Overpass API — 5,981 real substations                              | \$12      |
| G4 | NEW      | R6b Network Topology modifier â centrality + ring analysis from OSM graph            | \$5       |
| G5 | NEW      | Network & Topology layer (H): 7 variables — OCCTO Grid Development Plan, ring analysis | \$12      |
| G6 | NEW      | Output Scores layer (I): 7 variables â risk_score, ETTC, stationary probs            | \$12      |

SECTION D — MONTHLY DEEP-DIVE

The Osaka–Aichi Corridor: Where Hydroelectric Dominance Meets Grid Modernization

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Osaka–Aichi corridor · **002** DER saturation hotspots (Tokyo/Fukuoka) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Nuclear fleet dependency and transition risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Prefecture Grid Health Cards · **010** Academic validation and peer feedback · **011** Asia–Pacific replication feasibility · **012** Year in review.

## D.1 Why the Osaka–Aichi, Why Now

OSAKA–AICHI SUBSTATIONS

**1408**

165 EHV · 1243 HV

HIGH BAND

**1146 + 60**

85.7% of corridor

CORRIDOR MEDIAN R

**0.584**

Fleet median: 0.570

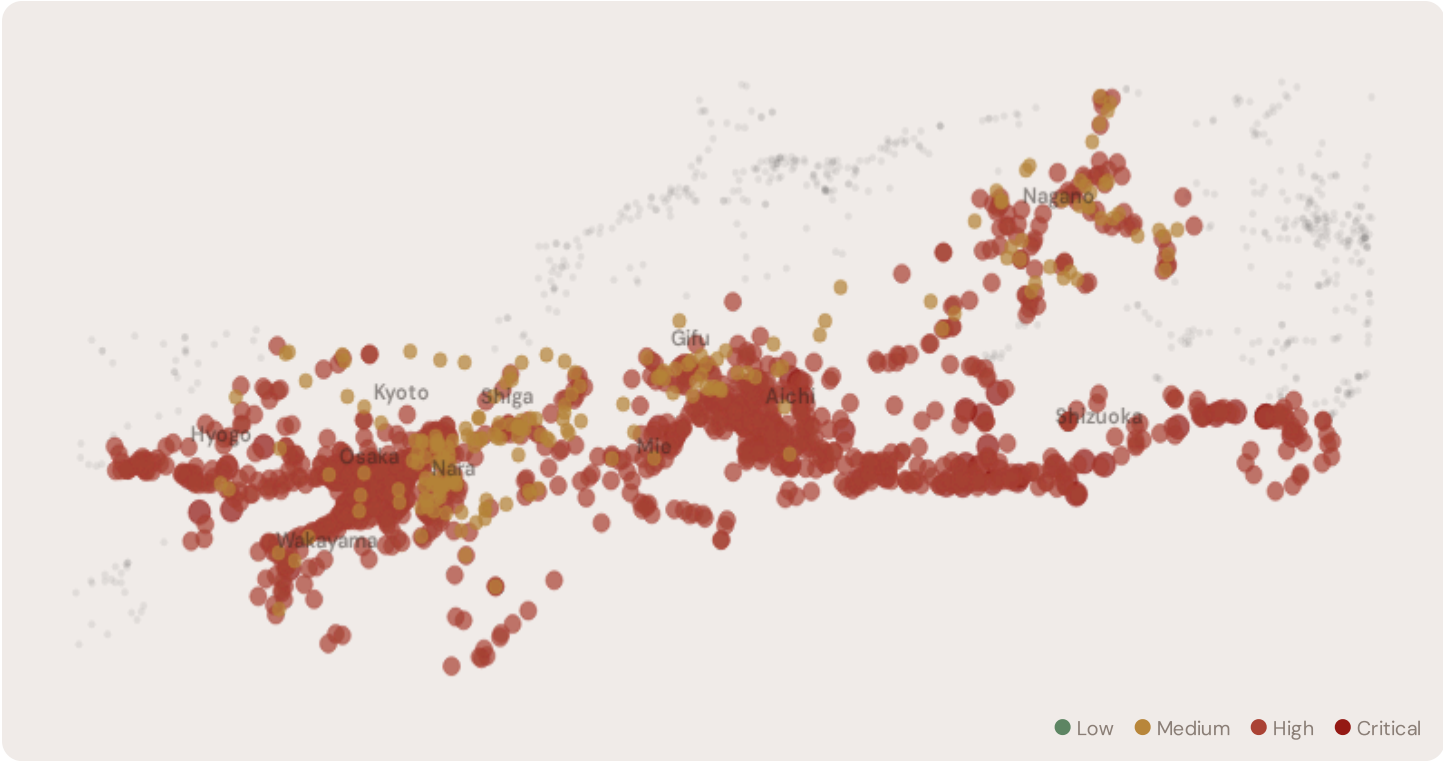
WORST PREFECTURE

**Shizuoka**

Avg R: 0.654 · 166 substations

The Osaka–Aichi hosts **1408 substations** across six prefectures, making it one of the most energy-critical corridors in Asia. However, its risk profile is disproportionately concentrated in the upper bands: **1146 substations (81.4%)** are classified High and **60 Critical**, with only **0** in the Low band. 56% of Osaka–Aichi substations score above the national fleet median of 0.570. The corridor median R of 0.584 reflects the tension between nuclear baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Japan 2030 timelines.

## D.2 Corridor Map and Scoring



| SUBSTATION         | PREFECTURE | R_FINAL | BAND     | C     | V     | I     | E     | S     | T     | ALERT |
|--------------------|------------|---------|----------|-------|-------|-------|-------|-------|-------|-------|
| 宮島変電所              | Shizuoka   | 0.879   | Critical | 0.348 | 0.790 | 0.534 | 0.624 | 0.631 | 0.408 | V     |
| 袋井変電所              | Shizuoka   | 0.852   | Critical | 0.348 | 0.790 | 0.496 | 0.624 | 0.505 | 0.408 | V     |
|                    | Hyogo      | 0.851   | Critical | 0.332 | 0.948 | 0.443 | 0.597 | 0.602 | 0.432 | V     |
|                    | Shizuoka   | 0.842   | Critical | 0.348 | 0.790 | 0.488 | 0.624 | 0.512 | 0.408 | V     |
| 飯村変電所              | Aichi      | 0.837   | Critical | 0.332 | 1.000 | 0.382 | 0.532 | 0.645 | 0.530 | V     |
| Sun'en Substation  | Shizuoka   | 0.836   | Critical | 0.348 | 0.790 | 0.491 | 0.624 | 0.473 | 0.408 | V     |
| Tomida Substation  | Aichi      | 0.830   | Critical | 0.332 | 1.000 | 0.490 | 0.532 | 0.644 | 0.530 | V     |
| 海南変電所              | Wakayama   | 0.824   | Critical | 0.365 | 0.540 | 0.497 | 0.695 | 0.646 | 0.376 | E     |
|                    | Aichi      | 0.822   | Critical | 0.332 | 1.000 | 0.506 | 0.532 | 0.430 | 0.530 | V     |
| Inazawa Substation | Aichi      | 0.815   | Critical | 0.332 | 1.000 | 0.513 | 0.532 | 0.630 | 0.530 | V     |

... 1398 additional substations — [explore full corridor in Digital Twin →](#)

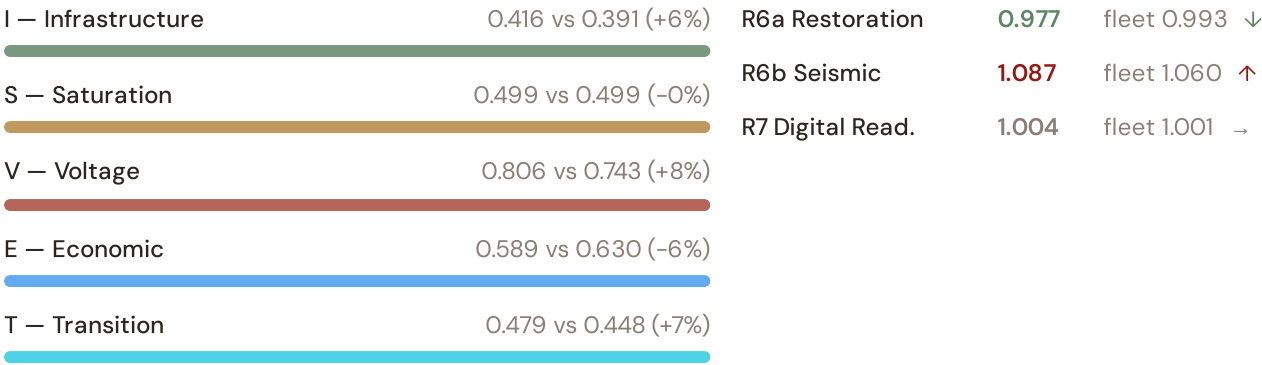
D.3 Component Analysis

COMPONENT AVERAGES — OSAKA-AICHI VS FLEET

C — Continuity 0.339 vs 0.371 (−9%)

MODIFIER AVERAGES — OSAKA-AICHI VS FLEET

R3 Consequence 1.112 fleet 1.109 ↑  
R4 Graph Crit. 1.060 fleet 1.054 ↑

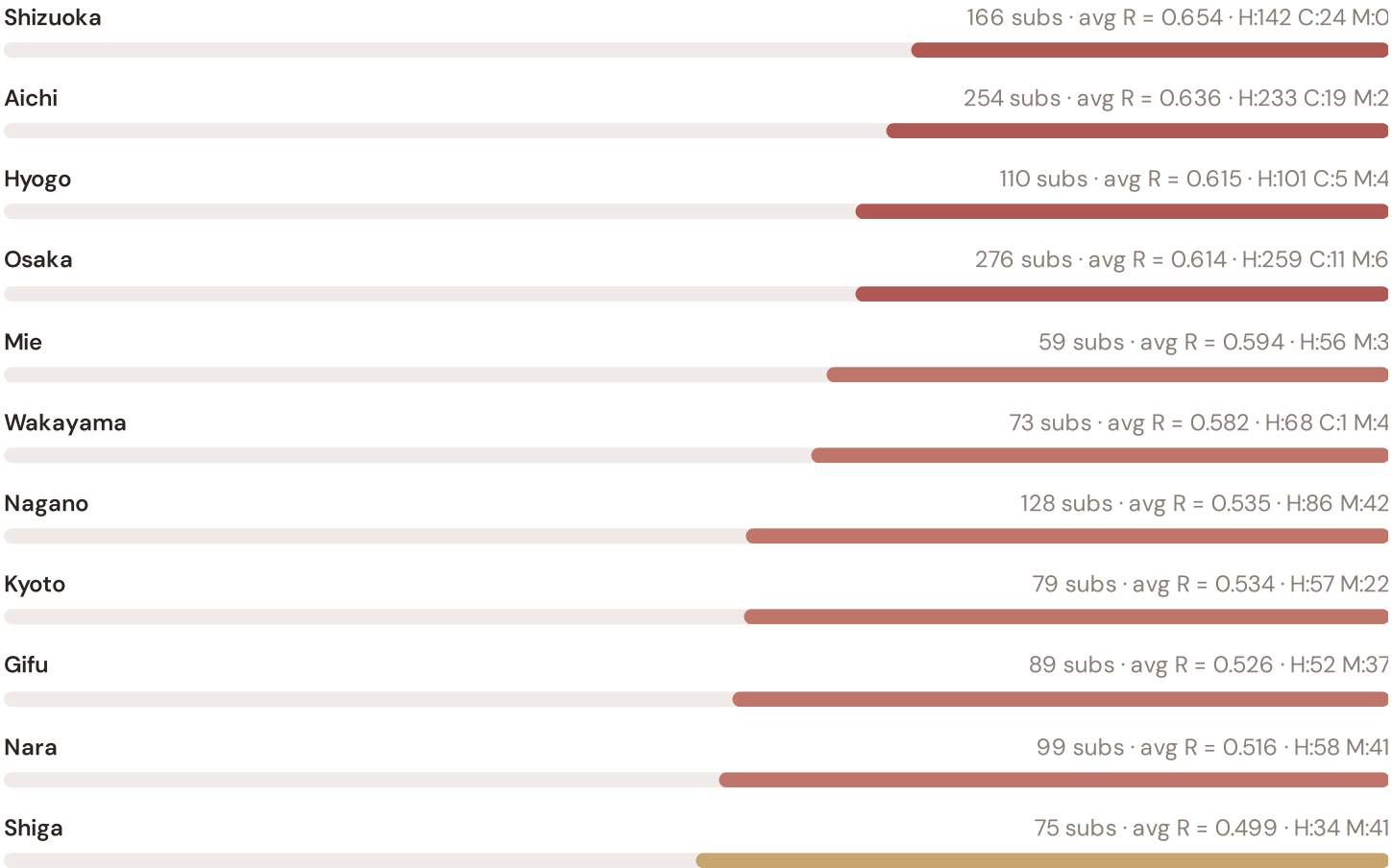


The dominant risk driver across the Osaka–Aichi corridor is **T (Transition)**, averaging 0.479 against a fleet average of 0.448 — occupying 957% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.806 vs fleet 0.743 (806% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.112 (fleet: 1.109), reflecting the socio-economic fragility of The Osaka–Aichi’s industrial-oriented, manufacturing-dependent economy with significant hydroelectric and nuclear bases. The R6b seismic modifier averages 1.087, capturing the region’s moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

## D.4 Prefecture Breakdown

PREFECTURE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The prefecture-level breakdown reveals significant intra-regional variation. **Shizuoka** leads with a mean R of 0.654 across 166 substations, while **Shiga** scores 0.499 — a spread of 0.156 within a single region. This disparity underscores why regional-level metrics (as used by OCCTO and IEEE) mask the true distribution of grid stress. The SSI’s substation-level granularity reveals that the Osaka–Aichi is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

## D.5 Implications

**314 Osaka–Aichi substations** score  $R \geq 0.650$ , placing them in the upper quartile of national risk. For OCCTO (Integrated Resource Planning) investment prioritisation, this analysis identifies the Shizuoka corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

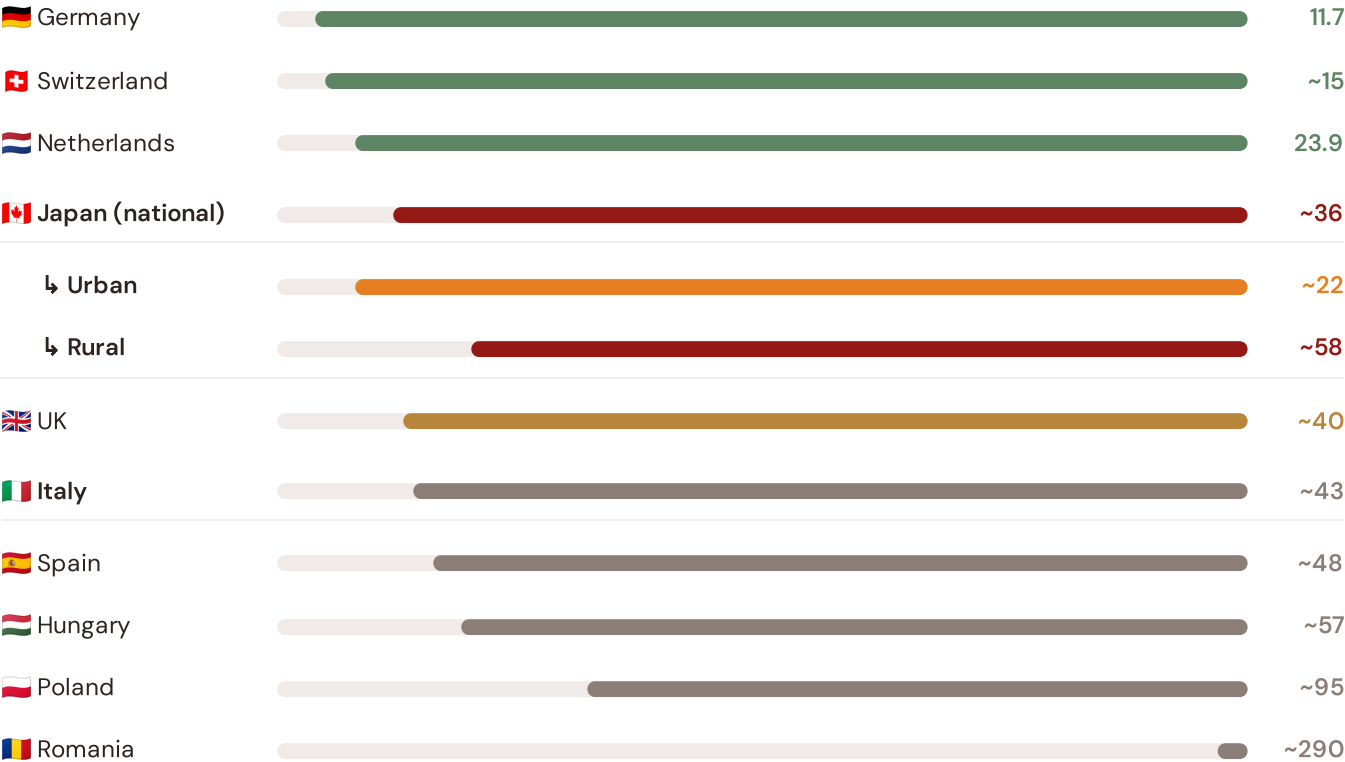
### SECTION E — RECURRING MODULE

## International Peer Benchmarking Card

**How this works:** Each edition includes one International Peer Benchmarking Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Osaka–Aichi corridor analysis hinges on Japan's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Asia-Pacific (003), CMIP6 climate hazard vs East Asian peers (004), etc. The underlying data updates annually with the OCCTO Benchmarking Report release.

### Unplanned SAIDI (min/customer/year) — Selected International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



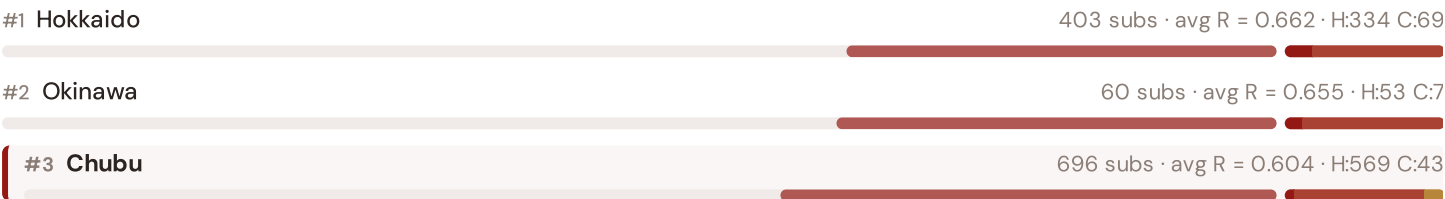
Source: OCCTO Reliability Report (2024); METI SAIDI/SAIFI (2023); ANRE / METI (2024). Japan sub-categories from METI territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with OCCTO BR release.

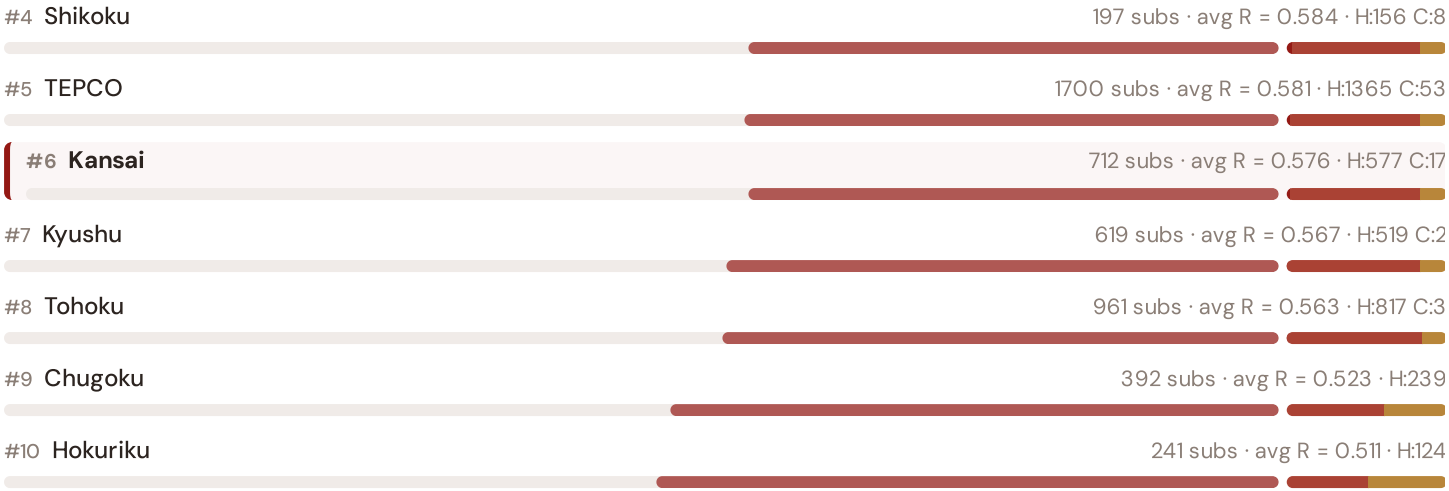
**This month's key insight:** The Osaka–Aichi corridor deep-dive shows **1408 substations** (of 1408) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Japan's rural territory classification. The Osaka–Aichi's average C of 0.339 is **0.9× the fleet average** of 0.371. In comparative terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~90–180 min/customer/year) than to Japan's own national average (~36 min). The SSI reveals what aggregate OCCTO statistics conceal: that Japan's mid-tier international position is not a national condition but a statistical average of top-tier prefectural performance and remote-territory-level rural performance — and the Osaka–Aichi corridor concentrates the worst of the latter.

### Regional Risk Ranking — All 10 EPCO Regions

Where does the Osaka–Aichi sit within the national landscape? Regions sorted by mean R<sub>median</sub>, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



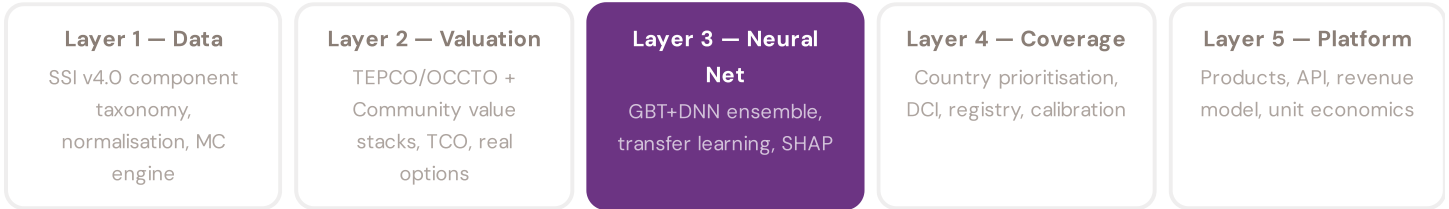


The Osaka–Aichi ranks **#3 of 47 prefectures** by mean  $R_{\text{median}}$ , placing it among the upper tier of national risk. The three most stressed regions are Hokkaido, Okinawa, Chubu, while the three most resilient are Hokuriku, Chugoku, Tohoku. 5 of 47 prefectures score above the fleet average of 0.578. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Japan. It is precisely this substation-level granularity that positions the SSI as a tool for OCCTO / Japan 2030 / OCCTO investment prioritisation: not treating entire prefectures as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

## SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the TEPCO/OCCTO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · Inaugural edition · April 2026

### Explainability & SHAP Values

*Clients need to understand not just the prediction but WHY a substation ranks highly*

Investment-grade models require interpretability. A BESS developer considering a ¥13M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends

primarily on the substation's poor reliability history — and can verify this against METI (Ministry of Economy, Trade and Industry)'s published SAIDI/SAIFI data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall visualisation: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Japanese training set comprises 5,981 substations with complete 95-feature vectors. Average CI\_width of 0.201 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 1026 Medium, 4753 High, 202 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

# Looking Ahead



#### NEXT MONTH — EDITION 002

### The BC Coastal Grid

Deep-dive into Hokkaido's grid risk profile — substation map, component analysis, prefecture breakdown. International Peer Benchmarking Card: Mediterranean thermal stress trajectory vs Atlantic peers.

#### NEXT DATA REFRESH

### Q3 2026 — 6 Quarterly Sources

**METI energy futures, EPCO utility filings, DER interconnection queues, Grid investment plans, OCCTO reliability assessments, SBJ business registry** are due for refresh in Q3. Impact: OSM (METI Asset Registry) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **SBJ Population Census / Current Population Survey** (4 variables → Population, elderly ratio, density, household counts by prefecture).

#### FLEET SPOTLIGHT

### o Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **O substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of May 2026. Deep-dive region: **Hokkaido**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Japanese utility, METI, OCCTO, TEPCO, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

## SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 5,981 substations · ~11,600 grid lines · 6 components · 20 metrics · 5 modifiers  
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5